

Amman Arab University

College of Information Technology — Department of Computer Science

Semester II – 2025/2026 | Course: 40742202 – Databases

Medical Clinic Management System

Database Project — Phase 2: Relational Schema

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5. Relational Schema

The following tables represent the complete relational schema mapped from the ER diagram. Each table specifies column names, data types, sizes, NULL constraints, domain constraints, and key designations.

Legend:

- PK — Primary Key
- FK — Foreign Key
- Alternating row fill for readability

Table: Doctor

Stores doctor profiles. Self-referencing FK for supervisor relationship.

Column Name	Data Type	Size	NULL?	Domain Constraint	Description	PK	FK	Note
doctorID	INT	-	NOT NULL	Positive integer	Unique doctor identifier	PK	-	-
firstName	VARCHAR	50	NOT NULL	Non-empty string	Doctor's first name	-	-	-
middleName	VARCHAR	50	NULL	String or NULL	Doctor's middle name	-	-	-
lastName	VARCHAR	50	NOT NULL	Non-empty string	Doctor's last name	-	-	-
specialization	VARCHAR	100	NOT NULL	Non-empty string	Medical specialization	-	-	-
phone_num	VARCHAR	20	NOT NULL	Digits, +, -, spaces only	Doctor's contact phone	-	-	-
superDocID	INT	-	NULL	Must exist in Doctor(doctorID) or NULL	Supervisor doctor ID (recursive)	-	Doctor(doctorID)	Self-ref

Table: Patient

Stores patient personal information.

Column Name	Data Type	Size	NULL?	Domain Constraint	Description	PK	FK	Note
patientID	INT	-	NOT NULL	Positive integer	Unique patient identifier	PK	-	-
firstName	VARCHAR	50	NOT NULL	Non-empty string	Patient's first name	-	-	-
middleName	VARCHAR	50	NULL	String or NULL	Patient's middle name	-	-	-

Column Name	Data Type	Size	NULL?	Domain Constraint	Description	PK	FK	Note
lastName	VARCHAR	50	NOT NULL	Non-empty string	Patient's last name	-	-	-
dateOfBirth	DATE	-	NOT NULL	Past date only	Patient's date of birth	-	-	-

Table: PatientAddresses

Multivalued attribute table — each row is one address for a patient. Composite PK: (patientID, Plocation).

Column Name	Data Type	Size	NULL?	Domain Constraint	Description	PK	FK	Note
patientID	INT	-	NOT NULL	Must exist in Patient(patientID)	FK to Patient	PK	Patient(patientID)	Composite PK
Plocation	VARCHAR	200	NOT NULL	Non-empty string	Patient's home address	PK	-	Composite PK

Table: Appointment

Represents M:N relationship between Doctor and Patient. Composite PK: (apptDate, apptTime, doctorID, patientID).

Column Name	Data Type	Size	NULL?	Domain Constraint	Description	PK	FK	Note
apptDate	DATE	-	NOT NULL	Valid calendar date	Appointment date	PK	-	Composite PK
apptTime	TIME	-	NOT NULL	Valid time value	Appointment time	PK	-	Composite PK
doctorID	INT	-	NOT NULL	Must exist in Doctor(doctorID)	Attending doctor	PK	Doctor(doctorID)	Composite PK
patientID	INT	-	NOT NULL	Must exist in Patient(patientID)	Attending patient	PK	Patient(patientID)	Composite PK

Table: Prescription

Each prescription is issued by one doctor to one patient.

Column Name	Data Type	Size	NULL?	Domain Constraint	Description	PK	FK	Note
Presc_number	INT	-	NOT NULL	Positive integer	Unique prescription ID	PK	-	-
issue_date	DATE	-	NOT NULL	Valid date, not future	Date prescription issued	-	-	-
patientID	INT	-	NOT NULL	Must exist in Patient(patientID)	Patient receiving Rx	-	Patient(patientID)	-
doctorID	INT	-	NOT NULL	Must exist in Doctor(doctorID)	Doctor who issued Rx	-	Doctor(doctorID)	-

Table: Medication

Catalog of medications. medName serves as PK.

Column Name	Data Type	Size	NULL?	Domain Constraint	Description	PK	FK	Note
medName	VARCHAR	100	NOT NULL	Non-empty string	Unique medication name	PK	-	-
TYPE	VARCHAR	50	NOT NULL	Non-empty string	Medication type/category	-	-	-

Table: Prescription_Medication

Junction table for M:N between Prescription and Medication. Composite PK: (presc_number, medName). Dosage is recorded per medication per prescription.

Column Name	Data Type	Size	NULL ?	Domain Constraint	Description	PK	FK	Note
presc_number	INT	-	NOT NULL	Must exist in Prescription(Presc_number)	FK to Prescription	PK	Prescription(Presc_number)	Composite PK
medName	VARCHAR	100	NOT NULL	Must exist in Medication(medName)	FK to Medication	PK	Medication(medName)	Composite PK
Dosage	VARCHAR	100	NOT NULL	Non-empty string e.g. '500mg twice/day'	Dosage instructions	-	-	-

6. Referential Integrity Constraints (Foreign Keys)

Doctor.superDocID	→	Doctor (doctorID)	Self-referencing: supervisor must be an existing doctor or NULL
PatientAddresses.patientID	→	Patient (patientID)	Each address belongs to an existing patient
Appointment.doctorID	→	Doctor (doctorID)	Appointment must reference a valid doctor
Appointment.patientID	→	Patient (patientID)	Appointment must reference a valid patient
Prescription.patientID	→	Patient (patientID)	Prescription must be for an existing patient
Prescription.doctorID	→	Doctor (doctorID)	Prescription must be issued by an existing doctor
Prescription_Medication.presc_number	→	Prescription (Presc_number)	Must reference a valid prescription
Prescription_Medication.medName	→	Medication (medName)	Must reference a valid medication